

Week 2 Assignment: A Unified Search Engine

BFS, DFS, UCS, and A-star behind one tested graph-search core

CMPSC 480 - Student Question Deck

Assessment overview

Implement and compare four search strategies while preserving one domain contract.

Coverage: Weeks 1-2: problem formulation, graph search, costs, heuristics, and testing

Points

10 total

Expected time

6-8 hours

Submission

Repository plus PDF report

Learning objectives

- ▶ Implement BFS, DFS, UCS, and A-star through a shared frontier abstraction.
- ▶ Use state keys and best-cost bookkeeping correctly in cyclic graphs.
- ▶ Justify admissibility and consistency of a grid heuristic.
- ▶ Compare strategies with solution-cost and search-effort measurements.
- ▶ Design edge-case tests that distinguish algorithmic guarantees.

Requirements checklist

- ▶ Use Python 3.11 and NumPy only unless the prompt explicitly permits another package.
- ▶ Preserve the published interfaces and make all tie-breaking deterministic.
- ▶ State assumptions, validate inputs, and handle empty, terminal, and unreachable cases.
- ▶ Include focused tests whose names explain the defect each test would expose.
- ▶ Report measured evidence; do not replace experimental results with unsupported claims.

Task 1 - Shared search contract (2 points)

Define the problem, node, result, and frontier interfaces before implementing a strategy.

- a. Specify the required fields and methods of SearchProblem, Node, and SearchResult. [1 points]
- b. Define an immutable state-key policy and list two input validation rules. [1 points]

Task 2 - BFS and DFS policies (2 points)

Implement breadth-first and depth-first graph search by changing only frontier removal order.

- a. Explain the exact frontier and explored-set behavior for BFS and DFS. [1 points]
- b. Give one test that separates BFS from DFS and one cyclic test that proves termination. [1 points]

Task 3 - UCS and A-star - part 1 (4 points)

Add cost-sensitive search and demonstrate that cheaper rediscovery replaces a worse frontier path.

- a. State the priority used by UCS and A-star and specify when the goal test occurs. [1 points]
- b. Trace a graph where A is first reached with cost 5 and later through B with cost 2. [1 points]
- c. Define Manhattan distance for a four-neighbor unit-cost grid and prove admissibility. [1 points]

Task 3 - UCS and A-star - part 2 (4 points)

Add cost-sensitive search and demonstrate that cheaper rediscovery replaces a worse frontier path.

- d. Prove consistency of Manhattan distance for each legal edge.

[1 points]

Task 4 - Experiments and defensive tests (2 points)

Evaluate all strategies on common instances and explain differences using theory.

- a. Report path cost, generated nodes, expanded nodes, and maximum frontier for at least three maps. [1 points]
- b. Include no-solution, single-cell, cheaper-rediscovery, and scale tests and state the bug each detects. [1 points]

Edge cases and defensive programming

- ▶ The initial state already satisfies the goal test.
- ▶ The graph contains a reachable cycle but no goal.
- ▶ A state is rediscovered through a strictly cheaper path.
- ▶ Two frontier entries have equal priority and require deterministic ties.
- ▶ The heuristic returns zero everywhere, reducing A-star to UCS.

Submission instructions

- ▶ Submit source, tests, and a README containing the exact test command.
- ▶ Include a PDF result table and a short explanation of every surprising measurement.
- ▶ Tag the final commit and ensure a clean clone reproduces the tests.

Do not submit credentials, generated caches, or unapproved libraries.

Grading rubric

Category	Points	Required evidence
Shared search contract	2	Typed interfaces, validation, and a deterministic result schema.
BFS and DFS policies	2	Working FIFO and LIFO policies plus trace tests.
UCS and A-star	4	Priority-frontier implementation, heuristic, and correctness traces.
Experiments and defensive tests	2	Reproducible result table, tests, and concise analysis.

Reflection prompt

In 150-200 words:

Identify one implementation choice that affected a theoretical guarantee. Cite a failing or passing test and explain what you changed.